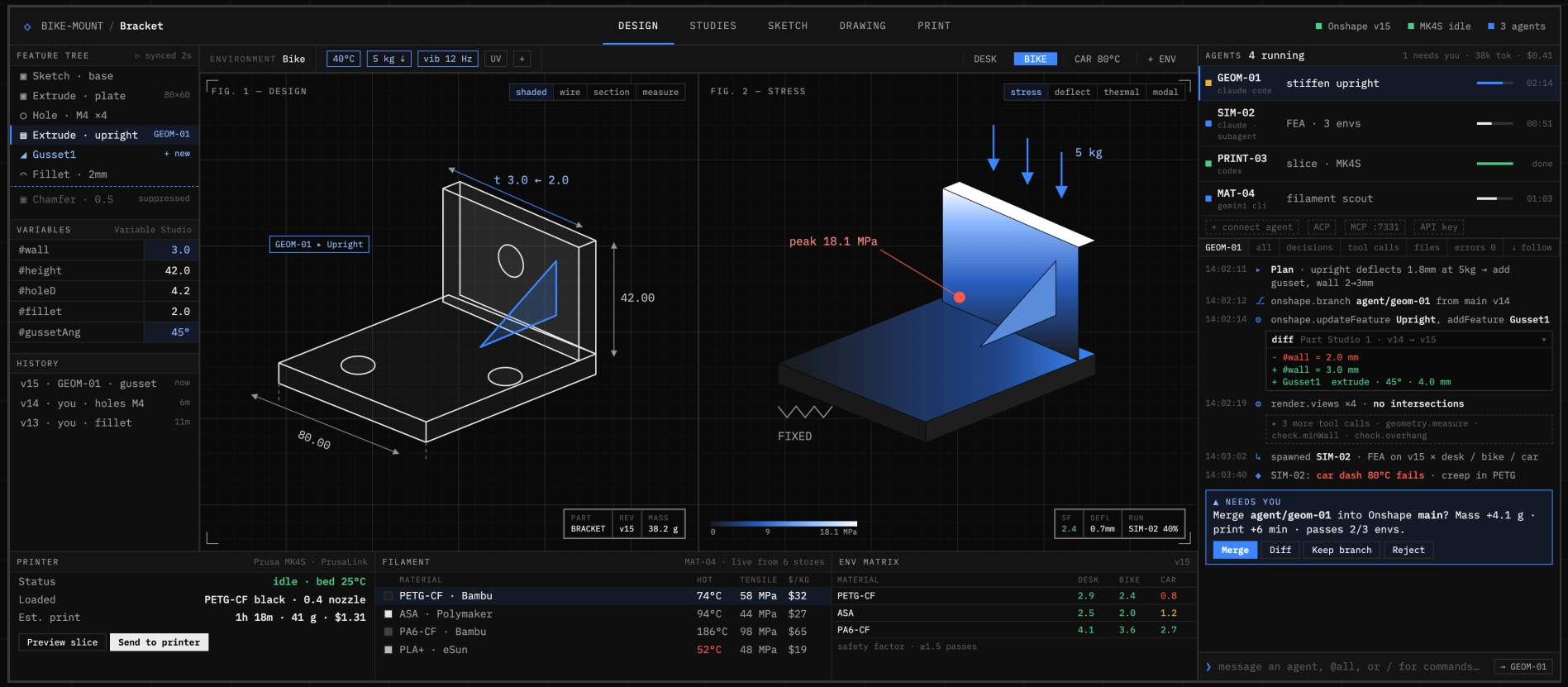
MISES

A desktop app where you and AI agents design physical parts together, on the same live model, test them in real-world conditions, and send them to your printer.



01 **Onshape stays the master.** Collaborate in the browser like always; agents work on their own branches.

02 **Any AI agent.** Claude, Codex, Gemini and others, several at once, all visible in one console.

03 **Test before you print.** A built-in simulator checks parts against heat, load and vibration.

The core loop: agent edits → test → you approve → print

Version 1 does one thing end to end. Everything else (assemblies, sketching, heavy studies) builds on top of this loop.

1	2	3	4	5	6
CONNECT	**ASK**	**ITERATE LOCALLY**	**TEST**	**APPROVE**	**PRINT**
Open an Onshape part. The tree, variables and 3D model mirror into the app.	Tell an agent what you want: "it flexes under 5 kg, stiffen it".	The agent tries variations on a copy on your computer. Free, fast, no Onshape calls.	The simulator checks each version against the environment you picked.	You see the change, the stress result and the cost, then merge into Onshape.	Slice and send to your Prusa or Bambu, with a final confirm.

WHY AGENTS WORK ON A LOCAL COPY

Onshape allows only 2,500 API calls a year on free and student plans, about 7 a day. If agents edited Onshape directly they would burn that in an afternoon.

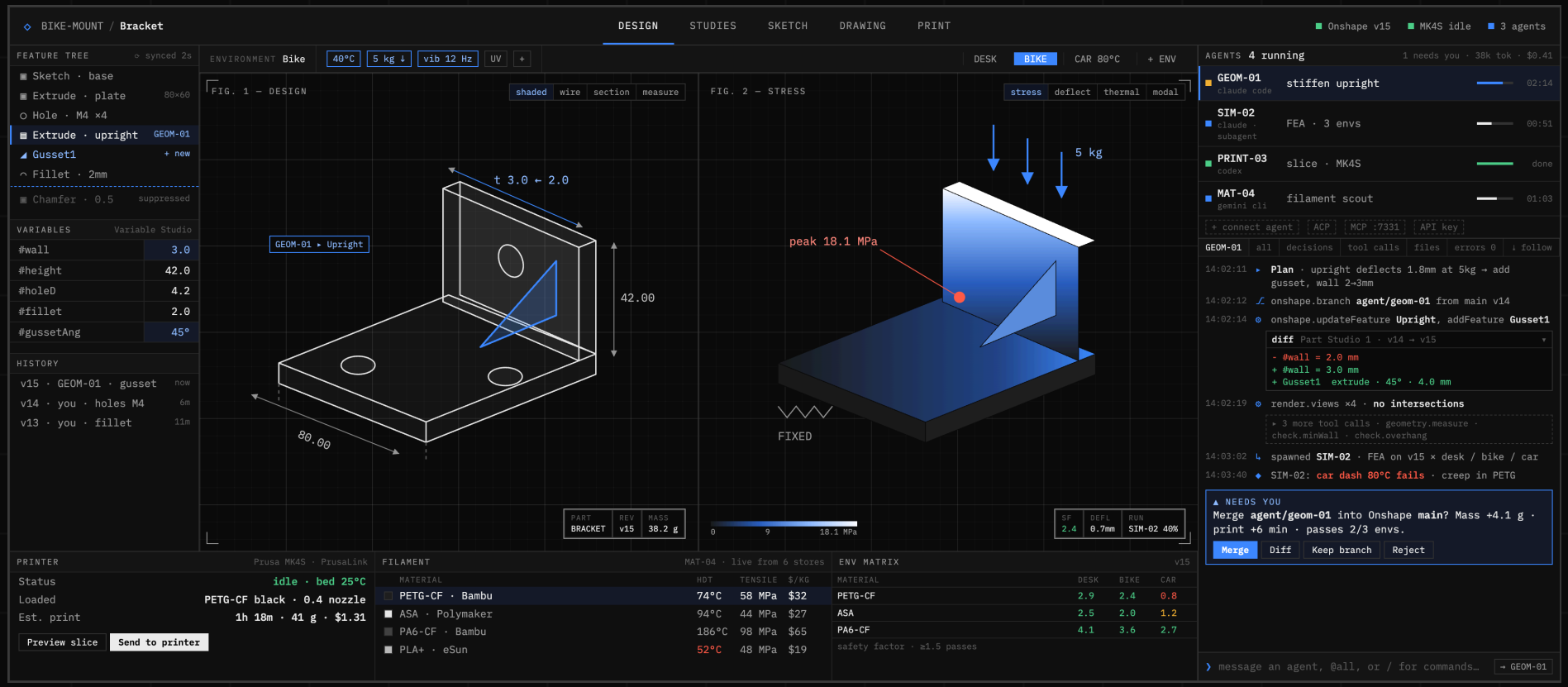
So the app keeps a simplified copy of each part, driven by the same Onshape variables. Agents experiment on it for free, and only the winning numbers go back to Onshape, about **3 calls per approved change**.

WHAT ALWAYS NEEDS YOU

- Merging a change into Onshape main
- Touching a face that mates with another part
- Letting an agent edit a part outside its scope
- Starting a print
- Spending paid credits

The simulator is a guide, not a guarantee. Every result says which solver and inputs it used.

The main screen



Top bar. Part name left, mode tabs center, status lights right for Onshape, printer and agents.

Feature tree + variables. Mirrored from Onshape. Values an agent changed glow blue.

Environment bar. Switch test conditions: desk, bike mount, car dashboard at 80°C, or your own.

Design | stress. Your model and its live stress map side by side, updating together.

Agent console. Every running agent, its steps, tool calls and diffs, with filters so lots of output stays manageable.

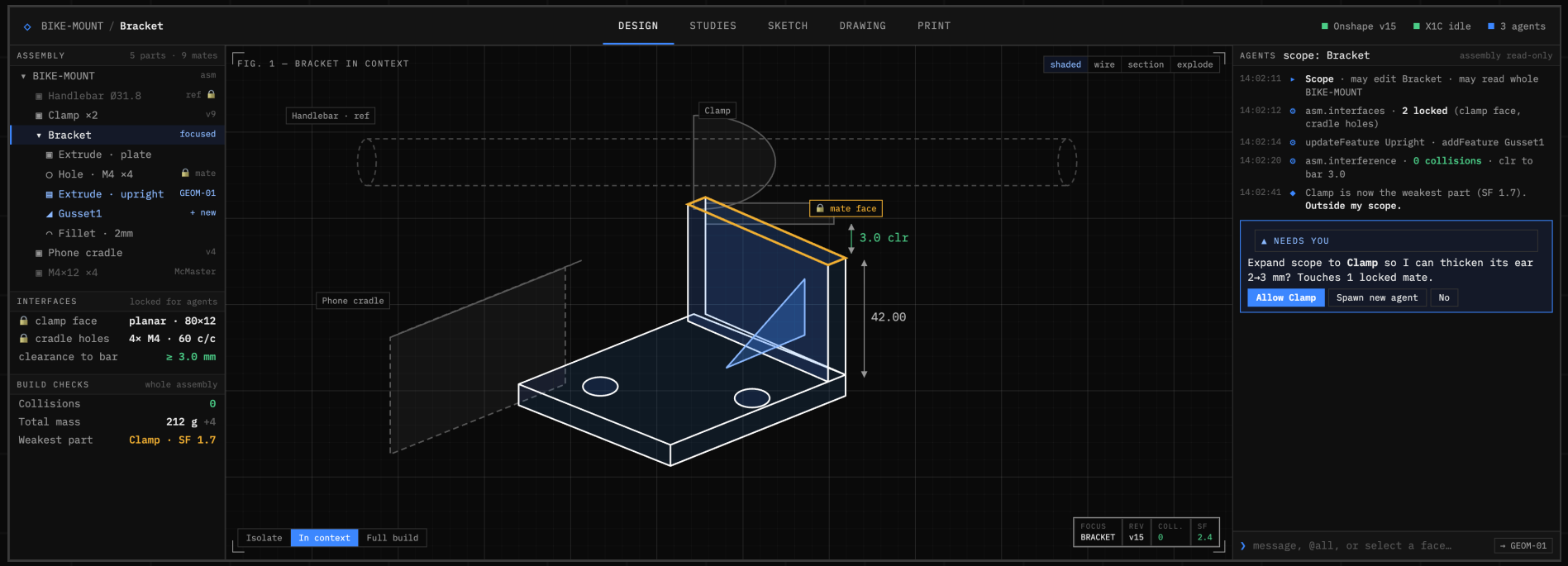
Approval card. Merge, diff, keep the branch, or reject, with mass, print time and test results.

Printer. Status, estimate, preview slice, send.

Filament. Live list with heat resistance, strength and price per kg.

Env matrix. Safety factor for each material in each environment. Red means it fails there.

Focus one part, keep the whole build in view



GHOSTED NEIGHBORS

The handlebar, clamp and phone cradle stay visible as grey outlines while you edit the bracket. Switch between Isolate, In context and Full build.

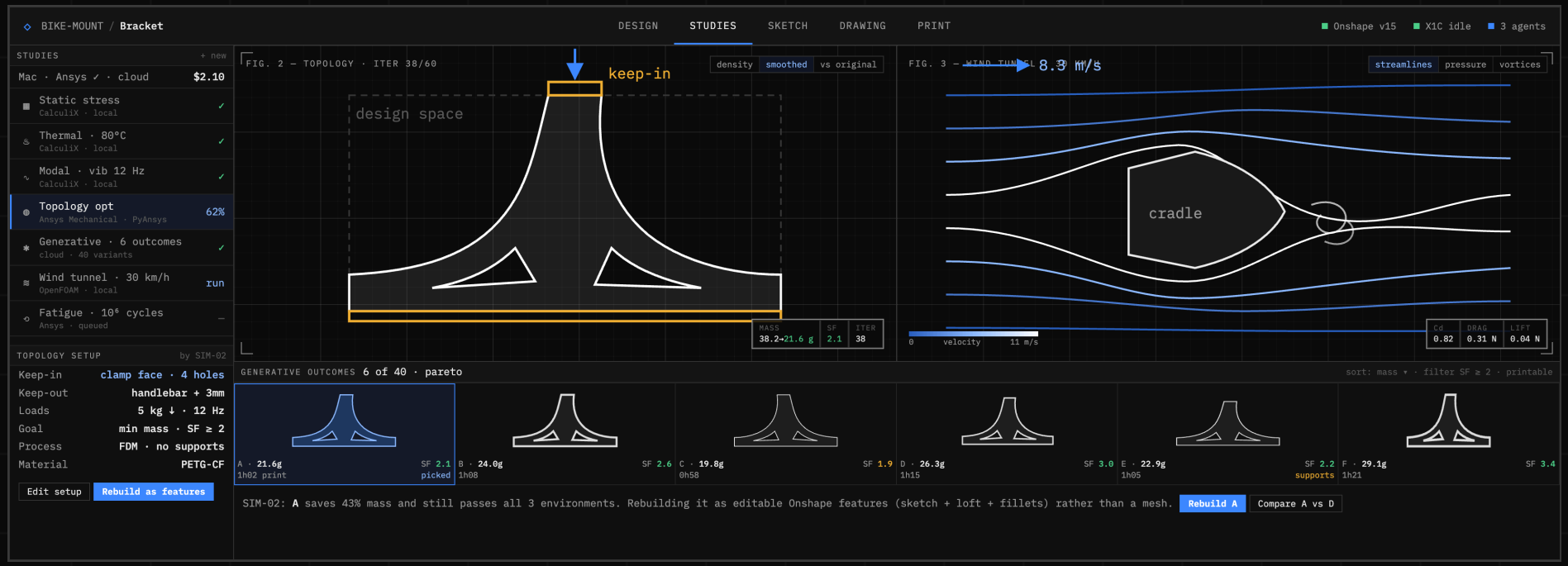
LOCKED INTERFACES

Faces that bolt or clamp to other parts are outlined amber and locked. Agents must ask before changing them.

WHOLE-BUILD CHECKS

Every change is checked for collisions and clearance. If the agent finds the weak point is in another part, it asks to expand its scope.

Topology optimization, generative design, wind tunnel



SOLVER PER STUDY

Light studies (stress, heat, vibration) run free on your computer. Heavy ones run in the cloud on the paid tier.

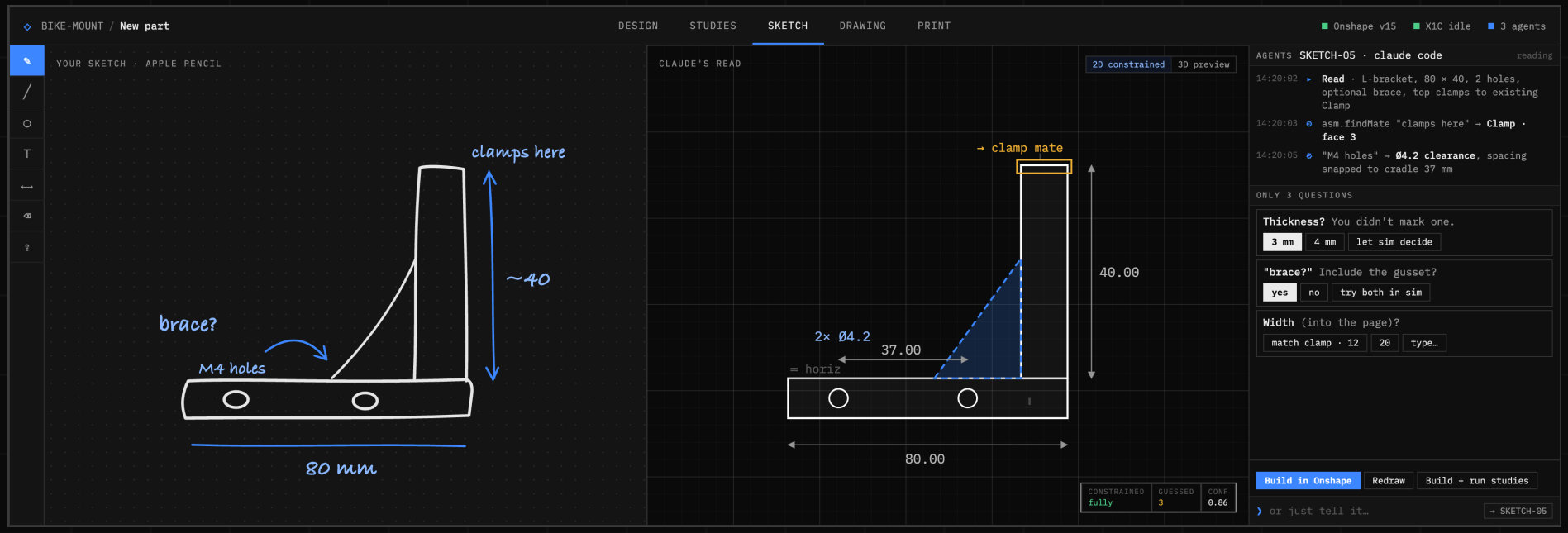
WEIGHT VS STRENGTH

Generative results line up as tradeoffs: mass, safety factor, print time, and warnings like "needs supports".

BACK TO REAL CAD

The chosen shape gets rebuilt as editable Onshape features, not a mesh. This starts rough, with your help, and gets more automatic later.

Draw it roughly, an agent builds it



ANY INPUT

Mouse, trackpad, pen tablet or Apple Pencil, or a photo of a napkin sketch. Scribbled numbers and notes count.

SHOWS ITS READING

Side by side: your sketch, and the clean dimensioned version the agent understood, with how much it had to guess.

FEW QUESTIONS

It links notes to the build ("clamps here" → the clamp's face) and asks only what it couldn't tell, then builds in Onshape.

UNDER THE HOOD

How it connects

One app for Mac and Windows. Everything runs on your computer by default; the cloud is only used for a small notification relay, accounts, and the paid heavy solvers.

THE APP WINDOW

Viewer, feature tree, agent console, simulator views, printer and filament panels.

LOCAL ENGINE (ON YOUR COMPUTER)

Does the real work.

`Onshape connection + call budget` `Local part copy` `Simulator` `Printer drivers` `Filament data`

ONE TOOL SERVER FOR EVERY AGENT

All agents use the same set of tools: edit the copy, run a test, propose to Onshape, slice, look up filament. Works with agents hosted in the app and with outside ones like Claude Code, Codex or Cursor.

`Claude` `Codex` `Gemini` `Cursor` `+ any compatible agent`

ONSHAPE
Master copy, collaboration, versions. Agents get their own branches.

YOUR PRINTER
Prusa (PrusaLink) and Bambu (LAN mode), on your network.

MISES CLOUD
Notification relay, accounts, paid heavy solvers.

BUILT TO OFFICIAL RULES

- Uses Anthropic's official Agent SDK and open agent standards, never leaked code.
- Users bring their own AI agent or key on the free tier, so AI costs are theirs.
- No tricks to dodge Onshape's limits; every call is counted and capped.

TO CONFIRM WITH ONSHAPE

If shared as a private app, every user's calls would count against the developer's allowance. Listing in Onshape's App Store should avoid that, but Onshape's own docs disagree with themselves, so this needs confirming before launch. Until then, users connect with their own Onshape keys.

PRICING AND PLAN

Cheap for everyone, paid for the power

FEATURE	FREE	PAID
Core loop with Onshape	✓	✓
Local stress, heat, vibration tests	✓	✓
Prusa + Bambu printing	✓	✓
Filament lookup	✓	✓
Bring your own AI agent	✓	✓
Agents at once	1–2	many
Topology optimization	—	✓ cloud
Generative design	—	✓ cloud
Wind tunnel (CFD)	—	✓ cloud
Hosted AI credits (no key needed)	—	✓
Team features	—	✓

BUILD ORDER

M0	App shell on Mac + Windows	M5	Prusa, then Bambu printing
M1	Onshape mirror + call budget	M6	Many agents at once
M2	Local part copy	M7	Assembly focus
M3	Agents + approve + merge	M8	Filament service
M4	Simulator	M9	Sketch to part
M10	Paid tier + cloud studies		

M0 to M5 is the full first version.

WHY THE FREE TIER STAYS CHEAP

Everything heavy runs on the user's own computer, AI costs are theirs, and Onshape calls hit their own allowance. Our running cost per free user is close to zero; the paid tier pays for cloud compute.